

Medical Image Processing

M2 Data Science & AI 2024-2025

www.sop.inria.fr/teams/asclepios/cours/MS_DS/

X. Pennec

Medical image registration



Epione team
2004, route des Lucioles B.P. 93
06902 Sophia Antipolis Cedex

<http://www.sop.inria.fr/epione>

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Medical Image Analysis – M2 DSAI 2024-2025

Thursday afternoon

Course notes : http://www.sop.inria.fr/teams/asclepios/cours/MS_DS/

- Thur. Jan. 09, 14:00-17:00 [Xavier Pennec] Introduction to Medical Image Acquisition
- Thur. Jan. 16 13:30-16:30 [Xavier Pennec] Medical Image Registration
- Thur. Jan. 23 13:30-16:30 [Hervé Delingette] Image Filtering
- Thur. Jan. 30 13:30-16:30 [Xavier Pennec] Statistics on Riemannian manifolds and Lie groups
- Thur. Feb. 6 13:30-16:30 [Hervé Delingette] Principles and Metrics for Image segmentation
- Thur. Feb. 13 13:30-16:30 [Hervé Delingette] Image segmentation based on clustering
- Thur. Feb. 27 13:30-16:30 [Xavier Pennec] Diffeomorphic Registration and Computational Anatomy
- Thur. March 6, 13:30-16:30 [Hervé Delingette] Connectivity and Shape Constrained Image segmentation
- Thur. March 13 2023 13:30-15:00, [XP & HD] EXAM

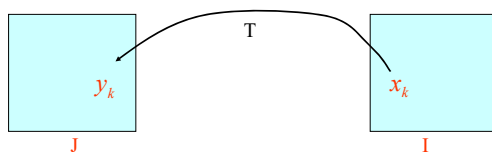
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Goals of Registration

A dual problem

- Find the point y of image J which is corresponding (homologous) to each points x of image I .
- Determine the best transformation T that superimposes homologous points



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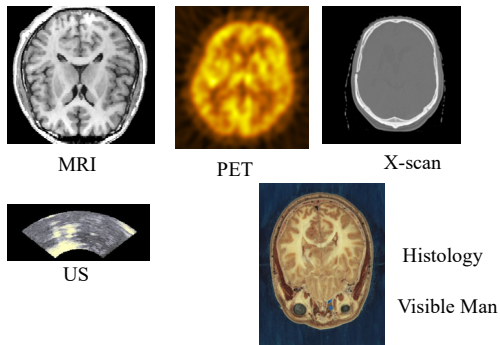
Principal Applications

- Fusion of multimodal images
- Temporal evolution of a pathology
- Inter-subject comparisons
- Superposition of an atlas
- Augmented reality

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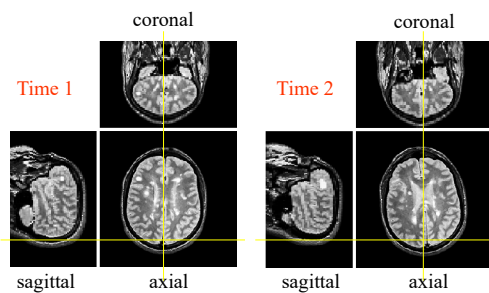
Fusion of Multimodal Images



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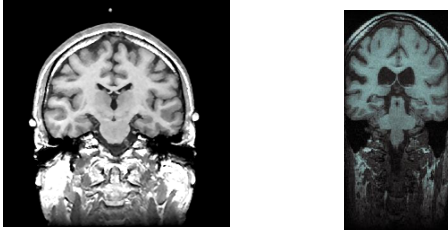
Temporal Evolution



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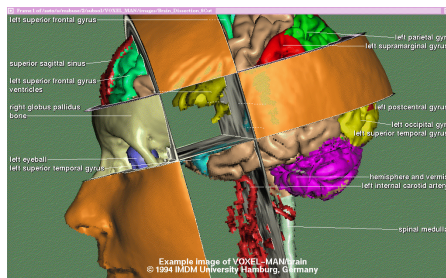
Inter-Subject comparison



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Registration to an Atlas



Voxel Man

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Augmented reality

Brigham & Women's Hospital



E. Grimson

Xavier Pennec

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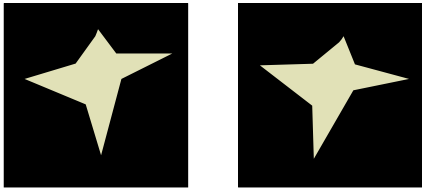
Classes of problems vs. applications

Temporal evolution	Intra Subject – Monomodal
Multimodal image fusion	Intra Subject - Multimodal
Inter-subject comparison	Inter Subject - Monomodal
Superposition on an atlas	Inter Subject - Multimodal

Intra Subject: Rigid or deformable
Inter Subject: deformable

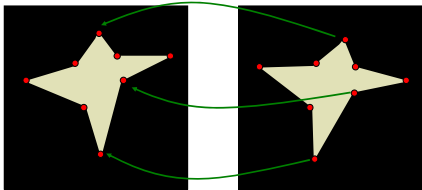
Intuitive Example

How to register these two images?



Feature-based/Image-based approach

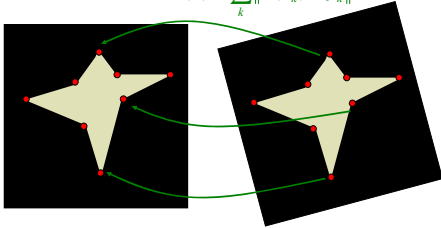
Feature detection (here, points of high curvature)
Measure: for instance $S(T) = \sum_k \|T(\mathbf{x}_k) - \mathbf{y}_k\|^2$



Feature-based/Image-based approach

Feature detection (here, points of high curvature)

Measure: for instance $S(T) = \sum_k \|T(\mathbf{x}_k) - \mathbf{y}_k\|^2$



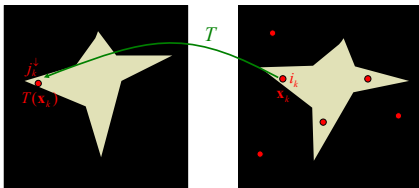
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Feature-based/Image-based approach

No segmentation!

Measure: e.g. $S(T) = \sum_k (i_k - j_k^\downarrow)^2$ Interpolation: $j_k^\downarrow = J(T(\mathbf{x}_k))$



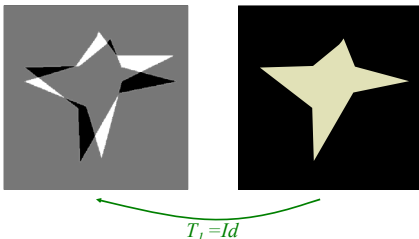
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Feature-based/Image-based approach

No segmentation!

Measure: e.g. $S(T) = \sum_k (i_k - j_k^\downarrow)^2$



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Feature-based/Image-based approach

No segmentation!

Measure: e.g. $S(T) = \sum (i_k - j_k^\downarrow)^2$

Partial overlap

T_2

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Classes of Transformations T

- ▢ Rigid (displacement)
- ▢ Similarities
- ▢ Affine (projective for 2D / 3D)
- ▢ Polynomials
- ▢ Splines
- ▢ Free-form deformations

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Classes of Transformations T

Rigid: $T(x) = Rx + t$

- ▢ Rotation and translation
- ▢ 6 parameters: (R: 3; t: 3)
- ▢ invariants: distances (isometry), frame orientation, curvatures, angles, straight lines

Similarities: $T(x) = s.Rx + t$

- ▢ Add a global scale factor
- ▢ 7 parameters
- ▢ invariants: ratio of distances, orientation, angles, straight lines

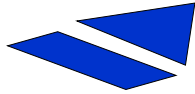
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Classes of Transformations T



Affine: $T(x) = Bx + t$

- 3x3 matrix B
- 12 parameters: (B: 9; t: 3)
- invariants: straight lines, parallelism



Quadratic: $T(x)^k = a_{ij}^k x_i x_j + b_i^k x_i + t^k$

- Add a symmetric 3x3 matrix A per axis
- 30 parameters (A: 18; B: 9; t: 3)
- invariants: do not preserve straight lines
any more

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Classes of Transformations T

Splines:

- Local polynomials of degree d, with a global continuity of degree C(d-1).
- number of parameters: depend on the number of control points (knots)
- locally affine: simplified version

Free form transformations: $T(x) = x + u(x)$

- a vector $u(x)$ is attached to each point x
- parameters: at most 3 times the number of voxels
- regularization: constrain to homeomorphisms (diffeomorphisms)



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Classification of registration problems

Type of transformation

- Parametric
Rigid (displacement), similarity, affine, projective
- Deformables
Polynomial, spline, free-form deformations

Type of acquisition

- Monomodal
- Multimodal

Homology of observed objects

- Intra-subject (generally a well posed problem)
- Inter-subject (one-to-one correspondences, regularization ?)

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Mathematical Formulation of registration (Brown, 1992)

■ **Registration:** Given two datasets (images) I and J , find the geometric transformation T that « best » aligns the physically homologous points (voxels)

$$\hat{T} = \arg \max_{T \in \mathcal{T}} S(I, J, T)$$

Optimization algorithm

Transformation space (rigid, affine, elastic,...)

Similarity measure

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Course overview

Feature-based registration

Multimodal Intensity-based Registration

Deformable intensity-based Registration

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Geometric methods

Extract geometric features

- Invariant by the chosen transformations
 - Points
 - Segments
 - Frames

Given two sets of features, registration consists in:

- Feature identification (similarity): Match homologous features
- Localization: Estimate the transformation T

Algorithms

- Interpretation trees
- Alignment
- Geometric Hashing
- ICP

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Artificial markers



Stereotactic frame

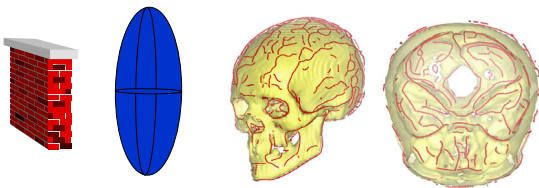
- Invasive
- External markers
- Motion
- Short time use

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Anatomical markers

Find geometric invariants to characterize a small number of singular points on anatomical surfaces



Generalization of edges and corner points to differentiable surfaces

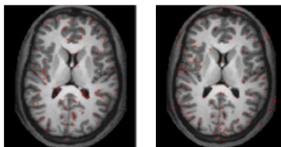
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Anatomical markers

Find geometric invariants to characterize a small number of singular points on anatomical surfaces

- Multiscale determinant of Hessian function (numerator of Gaussian curvature)
- 3D Harris detector [Rohr 99, Ruiz-Alzola et al 2001] based on the local correlation matrix $C = E(\nabla I \cdot \nabla I^T)$



Detected salient points in a axial slice of a brain. In a) Beaudet/Thirion curvature based detector, in b) the Harris/Rohr correlation based method is shown. [From Lloyd, Szekely, Kikinis, Warfield 2005]

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Iterative closest point (ICP)

One global criterion, $C(A, T) = \sum_i \text{dist}(T(P_i), A(P_i))^2$

alternatively minimized over

- Step 1: matches $A^* = \text{Argmin}_A \sum_i \text{dist}(T(P_i), A(P_i))^2$
- Step 2: transformation $T^* = \text{Argmin}_T \sum_i \text{dist}(T(P_i), A(P_i))^2$

Positive and decreasing criterion: convergence

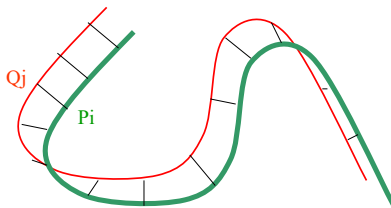
Robustness w.r.t. outliers: robust distances

$$\begin{aligned} \Phi(x) &= \|x\|^2 \quad (\text{standard mean}) \\ \text{dist}(x, y)^2 &= \Phi(\|x - y\|) \quad \Phi(x) = \min(\|x\|^2, \chi^2) \quad (\text{saturated mean}) \\ \Phi(x) &= \|x\| \quad (\text{median}) \end{aligned}$$

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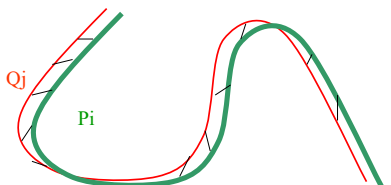
1. Optimization for matches



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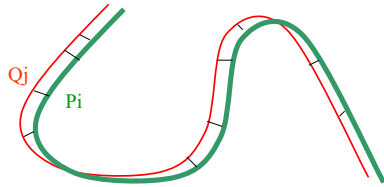
2. Optimization for T



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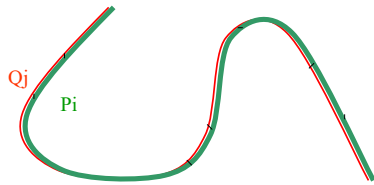
1-bis Optimization for matches



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2-bis Optimization for T



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Augmented reality guided radio-frequency tumor ablation

Collaboration with IRCAD (Strasbourg, France)

- Per-operative CT "guidance"
- Respiratory gating
- Marker based 3D/2D rigid registration



S. Nicolau, X.Pennec, A. Garcia, L. Soler, N. Ayache

IRCAD

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Increase 3D/2D registration accuracy: A new Extended Projective Point Criterion

Standard criterion:

$$\sum_{l=1}^M \sum_{i=1}^N \|P^l(T^* M_i) - m_i^l\|^2$$

- image space minimization (ISPPC)
- noise only on 2D data

Complete statistical assumptions + ML estimation

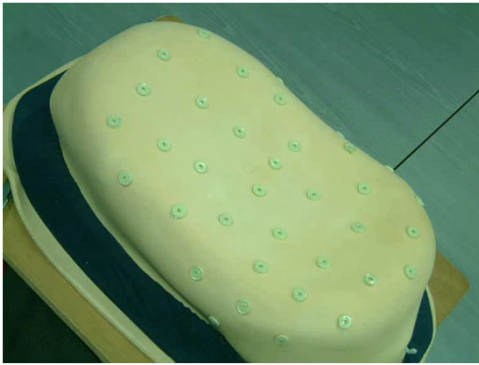
- Gaussian noise on 2D and 3D data
- Hidden variables M_i (exact 3D positions)

$$\sum_{l=1}^M \sum_{i=1}^N \frac{\|P^l(T^* M_i) - \tilde{m}_i^l\|^2}{2\sigma_{2D}^2} + \sum_{i=1}^N \frac{\|M_i - \tilde{M}_i\|^2}{2\sigma_{3D}^2}$$

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Registration for Augmented reality



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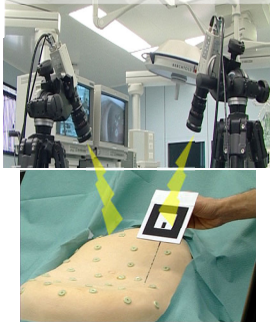
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Whole loop accuracy evaluation

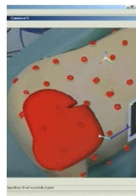
Left monitor with AR
& needle tracking



Stereoscopic HD video acquisition



Right monitor with AR
& needle tracking



Real scene: phantom + needle

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Whole loop accuracy evaluation

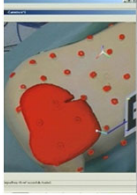
Left monitor with AR & needle tracking

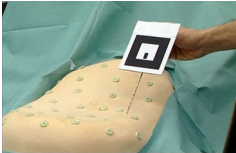




Endoscopic control

Right monitor with AR & needle tracking

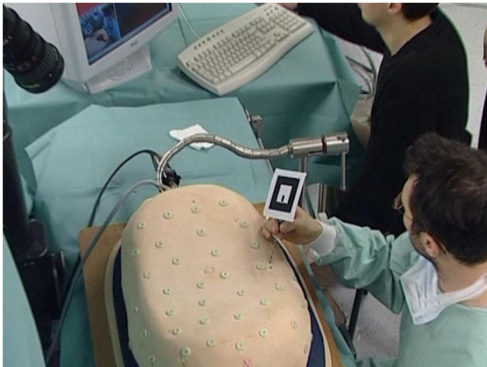




Real scene: phantom + needle

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Whole loop accuracy evaluation



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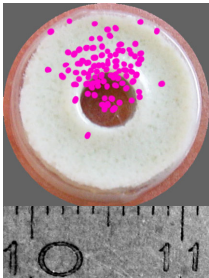
Whole loop accuracy evaluation

Experimental setup

- Two participants (comp. sci. + surgeon)
- 100 needle targeting

Measures

- Distribution of hits (endoscopic view, video recording)
- Average deviation from target: $2.8 \text{ mm} \pm 1.4$
- Average targeting time: $46.6 \text{ sec.} \pm 24.64$



[S. Nicolau, A. Garcia et al., Aug. & Virtual Reality Workshop, Geneva, 2003]

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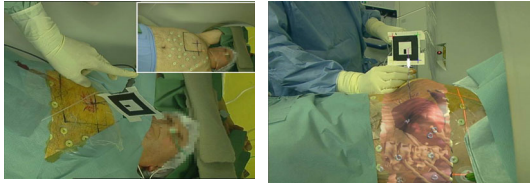
Liver puncture guidance using augmented reality

3D (CT) / 2D (Video) registration

- 2D-3D EM-ICP on fiducial markers
- Certified accuracy in real time

Validation

- Bronze standard (no gold-standard)
- Phantom in the operating room (2 mm)
- 10 Patient (passive mode): < 5mm (apnea)



[S. Nicolau, PhD'04 MICCAI05, ECCV04,, IS4TM03, Comp. Anim. & Virtual World 2005]

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Course overview

Feature-based registration

Multimodal Intensity-based Registration

Deformable intensity-based Registration

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Intensity-based methods

No geometric feature extraction

Advantages:

- Noisy images and/or low resolution
- Multimodal images

Drawbacks:

- All voxels must be taken into account

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Compare multi-modal images?

Which similarity criterion ?

- Many available criteria:
 - SSD, Correlation, Mutual Information...?
- Variable costs and performances
- Where is the optimum ?

Maintz & Viergever, Survey of Registration Methods, Medical Image Analysis 1997

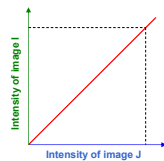
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Classification of existing measures

Assumed relationship

Intensity conservation



Adapted measures

Sum of Square Differences

Sum of Absolute value of Differences

Measures of intensity differences (Buzug, 97)

$$S(T) = \sum_k (i_k - j_k^I)^2$$



Interpolation: $j_k^I \equiv J(T(\mathbf{x}_k))$

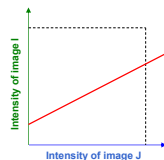
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Classification of existing measures

Assumed relationship

Affine



Adapted measures

Correlation coefficient

$$\rho_{IJ}(T) = \frac{1}{n\sigma_I\sigma_J} \sum_k (i_k - \bar{I})(j_k^I - \bar{J})$$

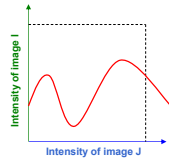
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Classification of existing measures

Assumed relationship

Functional



Adapted measures

Woods' criterion (1993)

Woods' variants (Ardekani, 95; Alpert, 96; Nikou, 97)

Correlation ratio (Roche, 98)

$$\eta^2 = \frac{\text{Var}[E(I | J(T))]}{\text{Var}(I)}$$

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Classification of existing measures

Assumed relationship

Statistical



Adapted measures

Joint Entropy (Hill, 95; Collignon, 95)

Mutual Information (Collignon, 95; Viola, 95)

Normalized Mutual Information (Studholme, 98)

$$MI(I, J) = H(I) + H(J) - H(I, J) = \sum_i \sum_j P(i, j) \log \frac{P(i, j)}{P(i)P(j)}$$

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A general framework

- A. Roche proposed a unifying maximum likelihood framework
- Physical and statistical modeling of the image acquisition process
- Create a hierarchy of criteria

A. Roche, G. Malandain and N. Ayache : *Unifying maximum likelihood approaches in medical image registration*. International Journal of Imaging Systems and Technology : Special Issue on 3D Imaging 11(1), 71-80, 2000.

- Based on pioneer works of (Costa et al, 1993), (Viola, 1995), (Leventon & Grimson, 1998), (Bansal et al, 1998)

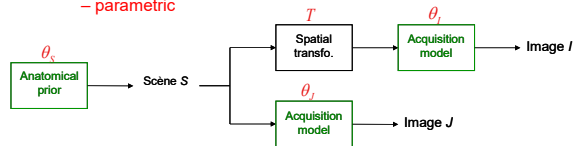
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Principle of the method [A. Roche]

Generative model of images

- probabilistic
- parametric



Maximum likelihood Inference

$$\max_{T, \theta_I, \theta_J, \theta_S} P(I, J | T, \theta_I, \theta_J, \theta_S)$$

Auxiliary variables: q

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Generic model

Scene: discrete random field (segmentation)



Assumptions:

- Spatial independence
- Stationarity

$$\Rightarrow P(S) = \prod_i \pi(s_i)$$

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Generic model

Images: noisy measures of the scene

$$\begin{cases} i_k &= f(s_k^\downarrow) + \varepsilon_k \\ j_l &= g(s_l) + \eta_l \end{cases} \quad s_k^\downarrow \equiv s(T(x_k))$$

Assumptions on ε and η :

- white (spatial indep.)
- Stationary
- Gaussian
- Additive
- Independent of each other

$$\Rightarrow P(I, J | S) = \prod_k G_{\sigma_\varepsilon}(i_k - f(s_k^\downarrow)) \times \prod_l G_{\sigma_\eta}(j_l - g(s_l))$$

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Likelihood function

Likelihood = joint law of images

$$P(I, J) = \int P(I, J | S) P(S) dS \\ \equiv L(T, \theta) \quad \text{with } \theta \equiv (\pi, f, g, \sigma_\epsilon, \sigma_\eta)$$

Log likelihood

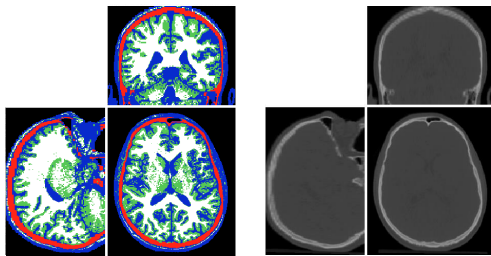
$$\log L(T, \theta) = \sum_{k \in \mathcal{A}} \log \frac{P_\theta(i_k, j_k^\downarrow)}{P_\theta(i_k) P_\theta(j_k^\downarrow)} + \sum_k \log P_\theta(i_k) + \sum_l \log P_\theta(j_l) \\ \text{with } P_\theta(i, j) = \sum_{p=1}^K \pi_p G_{\sigma_\epsilon}(i - f_p) G_{\sigma_\eta}(j - g_p)$$

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Example: X-Scan / MRI rigid registration

The estimation of θ allows the a posteriori estimation of the scene



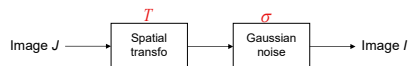
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Particular case 1

Assumptions

$$\begin{cases} f = g \\ |\eta| \ll |\epsilon| \end{cases} \Rightarrow i_k \approx j_k^\downarrow + \epsilon_k$$



Maximum Likelihood

$$\max_{\theta} L(T, \theta) \propto -n \log \left(\frac{c}{n} \sum_k (i_k - j_k^\downarrow)^2 \right) + E_{\text{cte}} \\ \text{SSD}$$

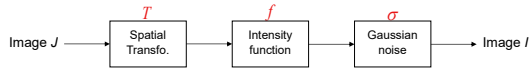
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Particular case 2

Assumptions

$$\begin{cases} g \text{ injective} \\ |\eta| \ll |\varepsilon| \end{cases} \Rightarrow i_k \approx f(j_k^\downarrow) + \varepsilon_k, \quad f \text{ constant by part}$$



Maximum Likelihood

$$\max_{\theta} L(T, \theta) \propto -n \log \left(\frac{c}{n} \min_f \sum_k (i_k - f(j_k^\downarrow))^2 \right) + E_{\text{cte}}$$

Squared fit error

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Particular case 2

Limit case: correlation ratio (CR)

$$\eta_{I|J}^2(T) = \frac{\min_f \sum_k (i_k - f(j_k^\downarrow))^2}{n \text{Var}(I)}$$

Squared fit error

Different normalization of the likelihood criteria

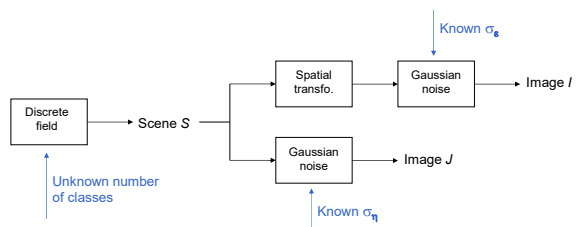
Equivalence ML / CR

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Particular case 3

Assumptions



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Particular case 3

Maximum Likelihood

$$\hat{P}(i, j) \approx \frac{1}{n} \sum_k G_{\sigma_t}(i - i_k) G_{\sigma_n}(j - j_k^\downarrow) \quad (\text{Parzen})$$

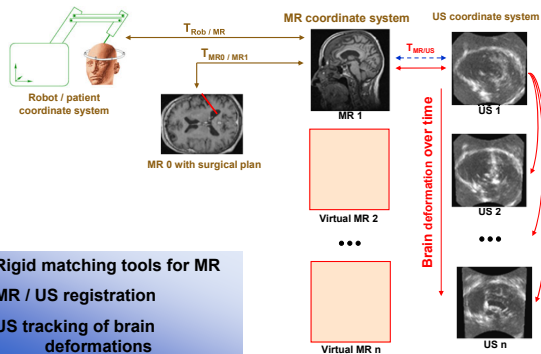
$$\max_P L(T, P) \propto \sum_k \log \frac{\hat{P}(i_k, j_k^\downarrow)}{\hat{P}(i_k) \hat{P}(j_k^\downarrow)} \quad \text{corrective terms}$$

$n \times \text{Mutual Information}$

Choice of a criterion

- Choosing a criterion imposes to deeply understand the physical image acquisition process
- Whenever several models are known, choosing the one with the smallest number of degrees of freedom increase the robustness of the approach.
- Current trend: learn it.
Pitfall: no idea of #dof and local minima

Roboscope



- Rigid matching tools for MR
- MR / US registration
- US tracking of brain deformations

Manipulator

Steady Hand Motion Compensation
Active Motion Constraints



Courtesy B. Davies & S. Starkie

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Manipulator

Steady Hand Motion Compensation
Active Motion Constraints



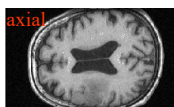
Courtesy B. Davies & S. Starkie

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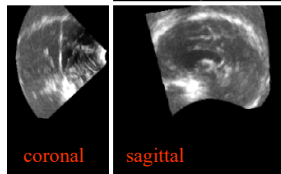
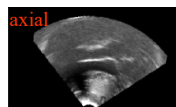
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MR-US Images

Pre - Operative MR Image



Per - Operative US Image



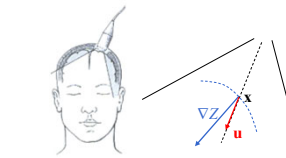
Acquisition of images : L. & D. Auer, M. Rudolf

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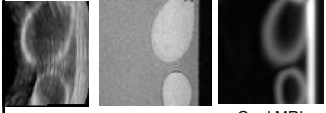
Ultrasound image / MRI registration

Elementary principles of US imagery



$$I_{rf}(\mathbf{x}) = |\nabla Z \cdot \mathbf{u}(\mathbf{x})| \times \xi(\mathbf{x})$$

Logarithmic compression

$$I(\mathbf{x}) \approx A \log |\nabla Z \cdot \mathbf{u}(\mathbf{x})| + B + \varepsilon(\mathbf{x})$$


US MRI Grad MRI

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Ultrasound image / MRI registration

Assumption: acoustic impedance is a function of the MR signal (denote by J)

$$Z(\mathbf{x}) = g(J(\mathbf{x})) \Rightarrow \nabla Z(\mathbf{x}) = g'(J(\mathbf{x})) \times \nabla J(\mathbf{x})$$

Relation between US and MR signals

$$I(\mathbf{x}) = f[J(\mathbf{x}), |\nabla Z \cdot \mathbf{u}(\mathbf{x})|] + \varepsilon(\mathbf{x})$$

In practice, the influence of orientation is neglected

$$I(\mathbf{x}) \approx f[J(\mathbf{x}), \|\nabla J(\mathbf{x})\|] + \varepsilon(\mathbf{x})$$

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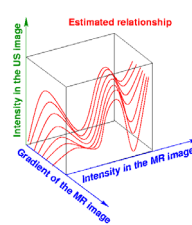
Bivariate Correlation Ratio

Intensity = function of 2 variables

$$I = f(J, \|\nabla J\|)$$

2 iterated stages

- Robust polynomial approx. of f
- Estimation of T:

$$\hat{T} = \arg \min_T \sum_k (I(x_k) - \hat{f}(J(T(x_k)), \|\nabla J(T(x_k))\|))^2$$


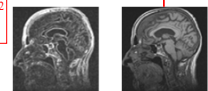
Estimated relationship

Intensity in the US image

Gradient of the MR image

Intensity in the MR image

Dependency Hypothesis



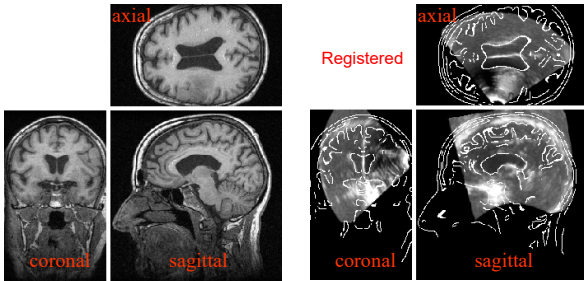
A. Roche, X. Pennec, G. Malandain, and N. Ayache
Rigid Registration of 3D Ultrasound with MR Images:
a New Approach Combining Intensity and Gradient Information.
IEEE Transactions on Medical Imaging, 20(10):1038--1049, October 2001.

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Typical Registration Result with Bivariate Correlation Ratio

Pre - Operative MR Image

Per - Operative US Image

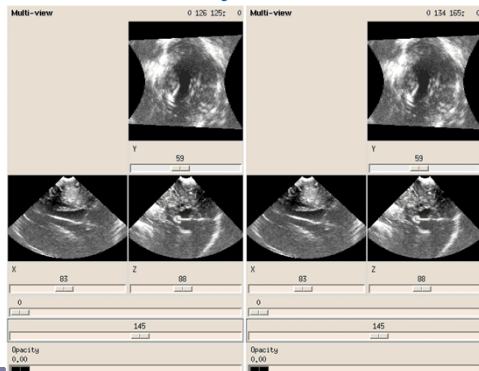


Acquisition of images : L. & D. Auer, M. Rudolf

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US Intensity MR Intensity and Gradient



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Course overview

Feature-based registration

Multimodal Intensity-based Registration

Deformable intensity-based Registration

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The deformable Registration Landscape in 1995

Optical flow

- Horn and Schunck, Artif. Intell. 17, 1981;
- Aggarwal and Nandhakumar, Proc. IEEE 76: 917–935, 1988;
- Barron *et al.*, 1994.

Linear elastic deformation

- Broit, PhD 1981.
- Bajcsy and Kovacic CVGIP 46, 1989
- Gee, Reivich, Bajcsy, *J. Comp. Assis. Tom.* 17, 1993.

Fluid (images & surface)

- Christensen, Rabbitt, Miller, *Phys. Med. Biol.* 39, 1994.
- Christensen, Rabbitt, Miller. IEEE Trans. Im. Proc. 5(10), 1996.
- Thompson and Toga, IEEE TMI 15(4), 1996.

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Mechanical deformations

T is a deformation endowed by its displacement vector field:

$$x_i \mapsto T(x_i) = x_i + u(x_i)$$

Similarity measure is the SSD

$$C = \sum (I(x) - J(x + u(x)))^2$$

The differential of this energy is considered as a force:

$$F(x, u) = -(I(x) - J(x + u)) \nabla J(x + u) \quad (1)$$

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Mechanical deformations

The force **F** is applied to the image considered

- Either as a linear elastic material (Lamé Coef.)

$$\mu \nabla^2 u + (\mu + \lambda) \nabla (\operatorname{div}(u)) = F \quad (2)$$

- Or as a viscous fluid (Navier-Stokes, Viscosity Coef.)

$$\mu \nabla^2 v + (\mu + \lambda) \nabla (\operatorname{div}(v)) = F \quad (3)$$

$$\frac{\partial u}{\partial t} = v - (\nabla u) \cdot v \quad (4)$$

Equations (2) and (3) are iteratively solved with **F** computed by (1).
u is computed by integrating equation (4).

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Demons' algorithm (MRCAS 95, CVPR96, Media98)

- Inspired by Christensen & Miller's work
- Algorithm in $O(N)$
- 2 alternated steps
 - Image forces create a displacement field u_n (normalized optical flow)
 - Regularization of u_n by Gaussian filtering

J.P. Thirion: Image Matching as a diffusion process: an analogy with Maxwell's demons.
Medical Image Analysis 2(3), 242-260, 1998.

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Demons' algorithm (MRCAS 95, CVPR96, Media98)

- $T_0 = \text{Identity}$
- Correction field
$$C_{n+1} = \frac{I - J \circ T_n}{\|\nabla I\|^2 + (I - J \circ T_n)^2} \nabla I$$
- Regularization by Gaussian filtering

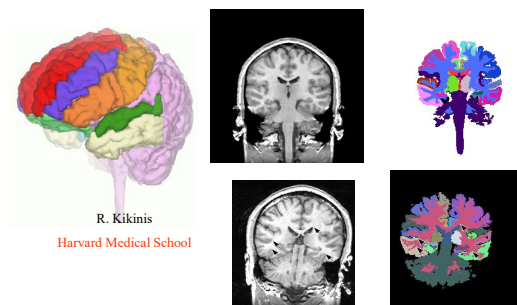
$$\begin{array}{ll} \text{Elastic} & \hat{C}_{n+1} = U_n \circ C_{n+1} \\ & U_{n+1} = G_\sigma * \hat{C}_{n+1} \\ \text{Fluid} & \tilde{C}_{n+1} = G_\sigma * C_{n+1} \\ & U_{n+1} = U_n \circ \tilde{C}_{n+1} \end{array}$$

J.P. Thirion: Image Matching as a diffusion process: an analogy with Maxwell's demons.
Medical Image Analysis 2(3), 242-260, 1998.

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Demons' algorithm (MRCAS 95, CVPR96, Media98)



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Intensity-based deformable registration

Demons algorithm: why does it work?

- + Fast, efficient
- - Do not minimize an energy
 - Difficult to analyze
 - Convergence?
 - Why does that work?
 - How to change the similarity measure?

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Course overview

Feature-based registration

Multimodal Intensity-based Registration

Deformable intensity-based Registration

- A historical perspective
- A Pair and Smooth approach
- Morphing

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Interpretation of demons

Why does that work? Convergence? Change the metric?

A variational framework to minimize a global energy

- Pennec-Cachier:

$$E = SSD + \int \|\nabla u\|_*^2$$

X. Pennec, P. Cachier and N. A. : Understanding the Demons Algorithm : 3D non rigid registration by gradient descent, MICCAI 1999, Springer-Verlag.

- Modersitzki: Min E with Neumann boundary conditions

$$E = SSD * + \int \|\nabla u\|^2$$

J. Modersitzki : Numerical Methods for Image Registration, Oxford University Press,2004.

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PASHA: Pair-And-Smooth, Hybrid energy based Algorithm

$$E(C, T) = \frac{1}{\sigma_i^2} SSD(I, J, C) + \frac{1}{\sigma_x^2} \|C - T\|^2 + \text{Reg}(T)$$

- **SSD** : measures the similarity of intensities
- **Reg** : regularization energy (quadratic)
- σ_x, σ_i : smoothing and noise parameters
- **C** : correspondences between points (vectors field)
- **T** : transformation (regularized vector field)
- Correspondences (matches) as an auxiliary variable

P. Cachier E. Babinet, E. Dormont, X. Pennec and N. A.: *Iconic Feature Based Nonrigid Registration: the PASHA Algorithm*, Comp. Vision and Image Understanding (CVIU), Special Issue on Non Rigid Registration, 89 (2-3), 272-298, 2003.

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PASHA: Pair-And-Smooth, Hybrid energy based Algorithm

$$E(C, T) = \frac{1}{\sigma_i^2} SSD(I, J, C) + \frac{1}{\sigma_x^2} \|C - T\|^2 + \text{Reg}(T)$$

Alternated minimization

- Minimization with respect to **C** :
 - Find matches between points by optimizing E_C + in the neighborhood of **T**
 - Gradient descent (1st, 2nd order, e.g. Gauss-Newton)
- Minimization with respect to **T** :
 - Find a smooth transformation that approximates **C**
 - Quadratic energy $\Rightarrow$ convolution
- **Interest**: fast computation

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Gauss-Newton optimization of the correspondences

$$E(C) = \int (I(x) - J(C(x)))^2 . dx + \frac{\sigma_i^2}{\sigma_x^2} \int \|C(x) - T(x)\|^2 . dx$$

Newton optimization

- Second order Taylor expansion of E(C)
- Hessian matrix can be null or negative

Gauss-Newton

- 1st order Taylor expansion of error
- $[I - J \circ (T + u)(x)] = [I - J \circ T(x)] + (\nabla J \circ T)^T . u(x) + O(\|u(x)\|^2)$
- Solve approximated SSD Criterion around C=T

$$E(C+u) \approx SSD(T) + 2 \int (J \circ T - I) . (\nabla J \circ T)^T . u + \int u^T . (\nabla J \circ T) . (\nabla J \circ T)^T . u + 2 \frac{\sigma_i^2}{\sigma_x^2} \int (C - T)^T . u + \frac{\sigma_i^2}{\sigma_x^2} \|u\|^2$$

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Gauss-Newton optimization of the correspondences

$$E(C) = \int (I(x) - J(C(x)))^2 dx + \frac{\sigma_I^2}{\sigma_x^2} \int \|C(x) - T(x)\|^2 dx$$

Exact solution of the quadratic approximation of the SSD

- Solve $\left[(\nabla J \circ T), (\nabla J \circ T)^t + \frac{\sigma_I^2}{\sigma_x^2} Id \right] u = (J \circ T - I), (\nabla J \circ T)$
- By inversion lemma: $u = \frac{(J \circ T - I), (\nabla J \circ T)}{\|\nabla J \circ T\|^2 + \sigma_I^2 / \sigma_x^2}$
- Local estimation of intensity variance: $\sigma_I^2 = (J \circ T - I)^2$
- Assuming isotropic voxel size: $\sigma_x^2 \approx 1$

$$u = \frac{I - J \circ T}{\|\nabla I\|^2 + (I - J \circ T)^2} \nabla I$$

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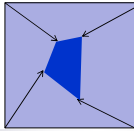
Important Practical Remark

$$u = \frac{I - J \circ T}{\|\nabla I\|^2 + (I - J \circ T)^2} \nabla I$$

- Norm of update is bounded by construction

$$(\|\nabla I\| - (I - J \circ T))^2 = \|\nabla I\|^2 + (I - J \circ T)^2 - 2(I - J \circ T)\|\nabla I\| > 0$$

$$\|u\| \leq 1/2$$
- Update is diffeomorphic by tri-linear interpolation!



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Efficient Regularization

Quadratic regularizer

$$\text{Reg}(T) = \int \sum_{k=1}^{\infty} \frac{\sum_{i_1 \dots i_k} \partial_{i_1} \dots \partial_{i_k} (T - Id)^2}{\sigma_d^{2k} k!}$$

Euler Lagrange optimization of $E(T) = \int \|C - T\|^2 + \text{Reg}(T)$

$$C - T + \sum_{k=1}^{\infty} \frac{(-1)^k \Delta^k (T - Id)}{\sigma_d^{2k} k!} = 0$$

Solution: Gaussian smoothing $T_{\text{opt}} = G_{\sigma} * C$ with $\sigma = 1 / \sigma_d$

Pennec, Cachier, Ayache, Understanding the "Demon's Algorithm": 3D Non-Rigid registration by Gradient Descent, MICCAI 1999.

Extension to a family of quadratic filters

$$G_{\sigma, \kappa}(\mathbf{u}) = \frac{1}{(\sigma \sqrt{2\pi})^3 (1 + \kappa)} \left(Id + \frac{\kappa}{\sigma^2} \mathbf{u} \mathbf{u}^T \right) \exp \left(-\frac{\mathbf{u}^T \mathbf{u}}{2\sigma^2} \right)$$

P. Cachier and N. Ayache, Isotropic energies, filters and splines for vectorial regularization. J. of Math. Imaging and Vision, 20(3):251-265, May 2004.

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Mixed Elastic / Fluid Regularization

$$E(C_n, T_n) = E_s(I, J, C_n) + \sigma \|C_n - T_n\|^2 + \cancel{\sigma \lambda \text{Reg}(T_n)} + \sigma \lambda [\omega \text{Reg}(T_n - T_{n-1}) + (1 - \omega) \text{Reg}(T_n)]$$

- Result is still obtained by convolution:

$$T_n = (1 - \omega) \cdot K^* C_n + \omega \cdot (T_n + K^*(C_n - T_{n-1}))$$

Advantages:

- Mixes fluid and elastic
- handles large displacements

P. Cachier N. A., *Isotropic Energies, Filters and Splines for Vector Field Regularization*, J. of Mathematical Imaging and Vision, 20: 251-265, 2004

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The Demons/PASHA Framework

Efficient energy minimization

$$E(C, T, \hat{T}) = \underbrace{E_s(I, J, C)}_{\text{similarity}} + \sigma \underbrace{\int \|C - T\|^2}_{\text{Auxiliary}} + \underbrace{\lambda \text{Reg}(T) + \mu \text{Reg}(\hat{T})}_{\text{Elastic + Fluid Regularity}}$$

Alternate Minimization

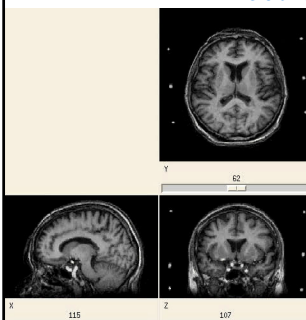
- on **C**, **Correspondance Field** (image forces)
Gauss-Newton gradient descent: normalized optical flow
- on **T**, **Deformation Field** (regularization)
Gaussian convolution

P. Cachier E. Bardinet, E. Dormont, X. Pennec and N. A.: *Iconic Feature Based Nonrigid Registration: the PASHA Algorithm*, Comp. Vision and Image Understanding (CVIU), 89 (2-3), 272-298, 2003.

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Inter-subject registration Affine transformation

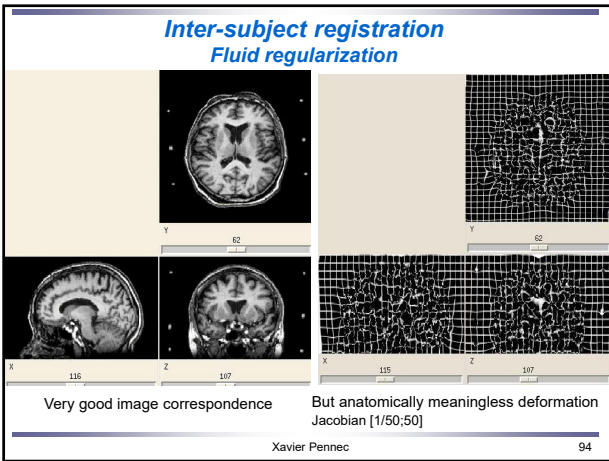


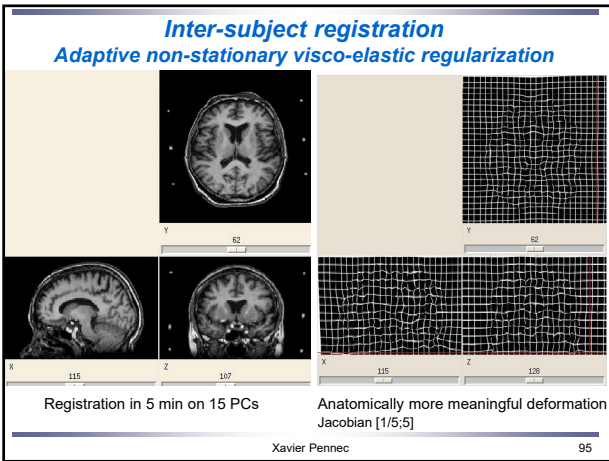
MR T1 Images
256x256x120 voxels
Atlas to patient registration
for radiotherapy planning

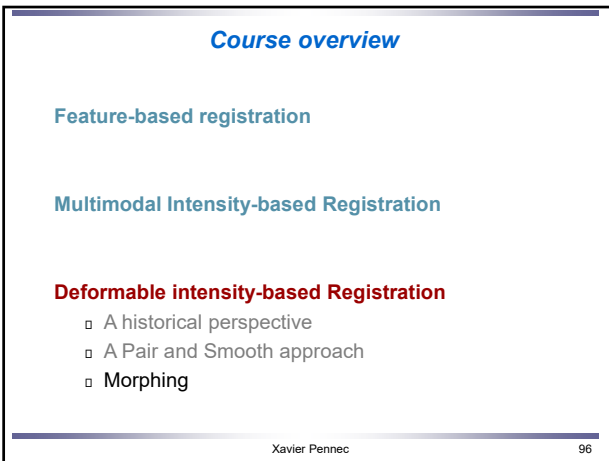
Correct size and position but high remaining variability in cortex and deep structures

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Using registration for image interpolation and morphing

Interpolating motion and intensity changes

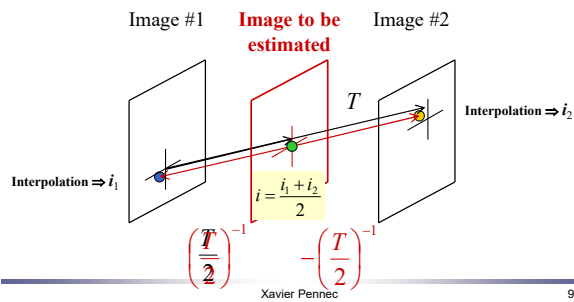
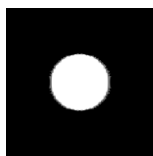


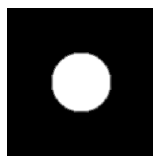
Image interpolation

Synthetic experiments

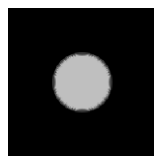
- Comparison with expected results



Translation



Shrinking

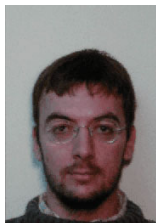
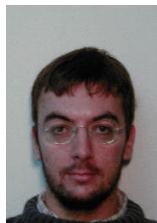


Shrinking + intensity
variation

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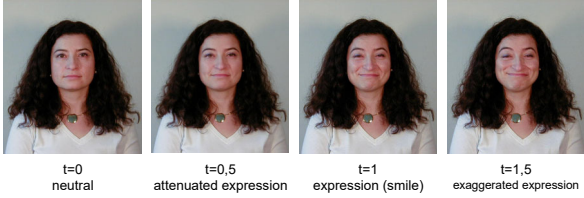
MORPHING



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Interpolation and Extrapolation



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